3 **(a)** **(i)** Explain the concept of *absolute zero* on the thermodynamic temperature scale.

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…….... [1]

(ii) Explain how empirical evidence leads to the concept of *absolute zero*.

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…….... [3]

(b) Two insulated gas cylinders A and B are connected by a tube of negligible volume, as shown in Fig. 3.1.

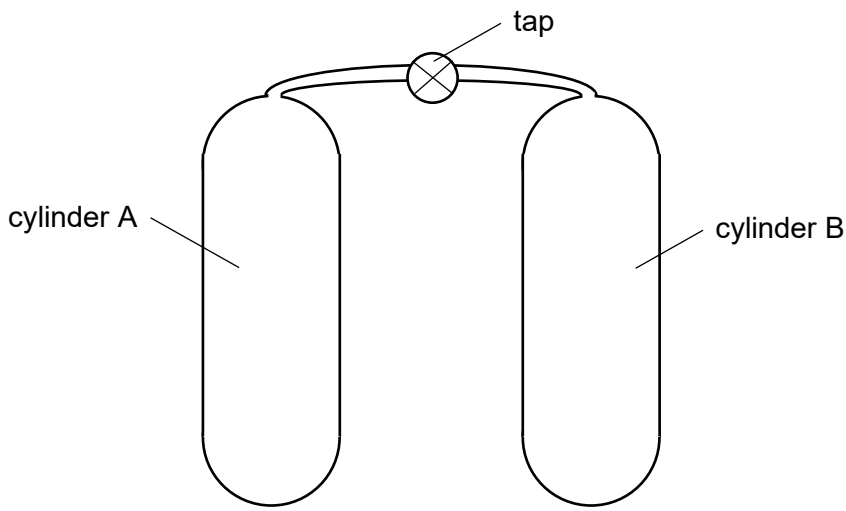

Fig. 3.1

Each cylinder has an internal volume of 2.0×10^{-2} m^3. Initially, the tap is closed and cylinder A contains 1.2 mol of an ideal gas at a temperature of 37 °C. Cylinder B contains the same ideal gas at pressure 1.2×10^5 Pa and temperature 37 °C

The tap is opened and some gas flows from cylinder A to cylinder B. Using the fact that the total amount of gas is constant, determine the final pressure of the gas in the cylinders.

pressure = …….. Pa [4]